

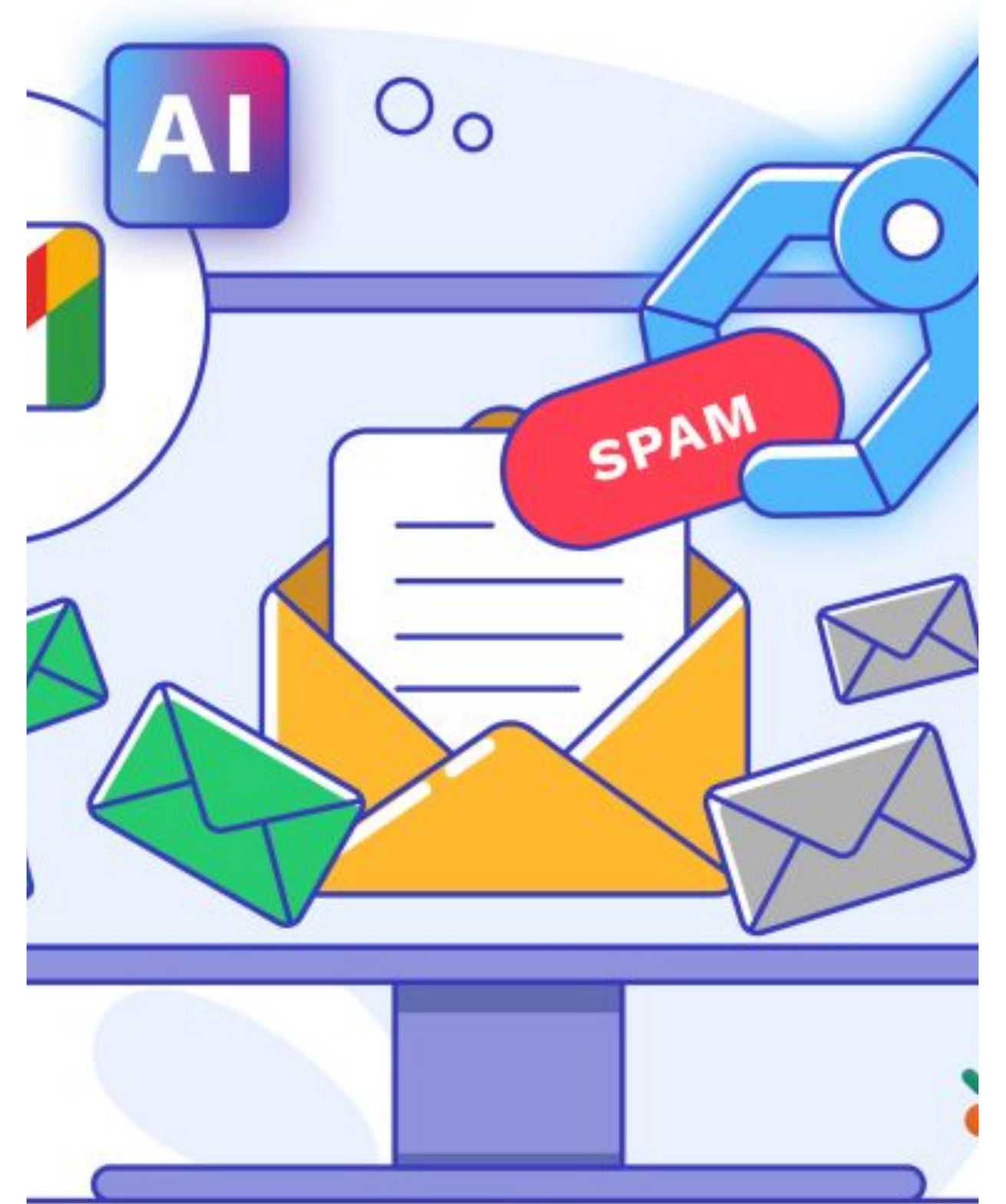
Email Spam Detection Progress Report

Group #7

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Overview of Goals

1

DATA PREPROCESSING & EDA

Understand the data and conduct
exploratory data analysis



DONE

2

MODEL DEVELOPMENT

Implement initial models and train models
using the training data



ONGOING

3

MODEL EVALUATION AND OPTIMIZATION

Evaluate model performance using test
data and experiment with different
algorithms and hyperparameters



NOT STARTED



Goal #1

Data Preprocessing & EDA



DONE

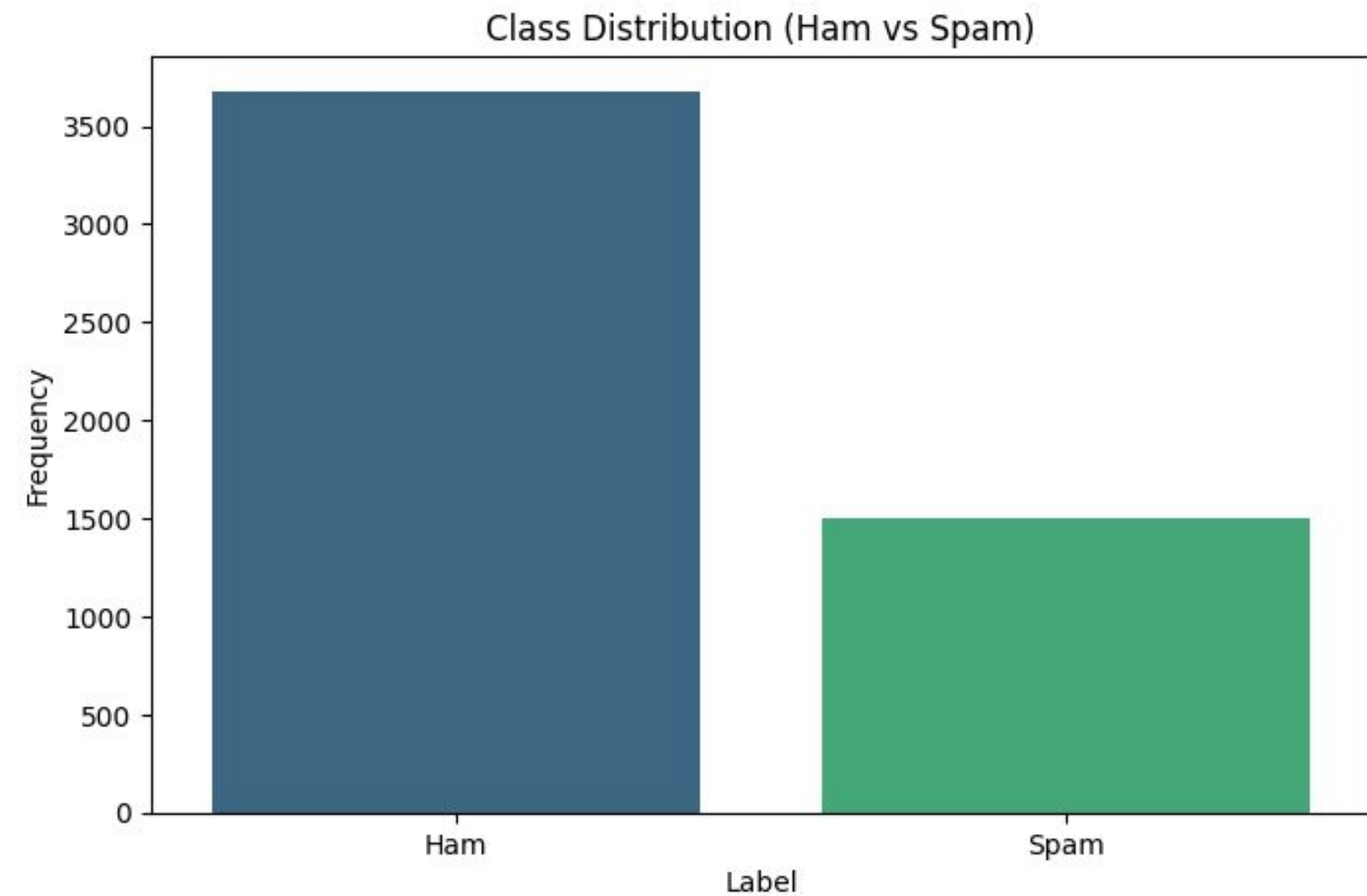
COLLABORATORS

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PROGRESS SUMMARY

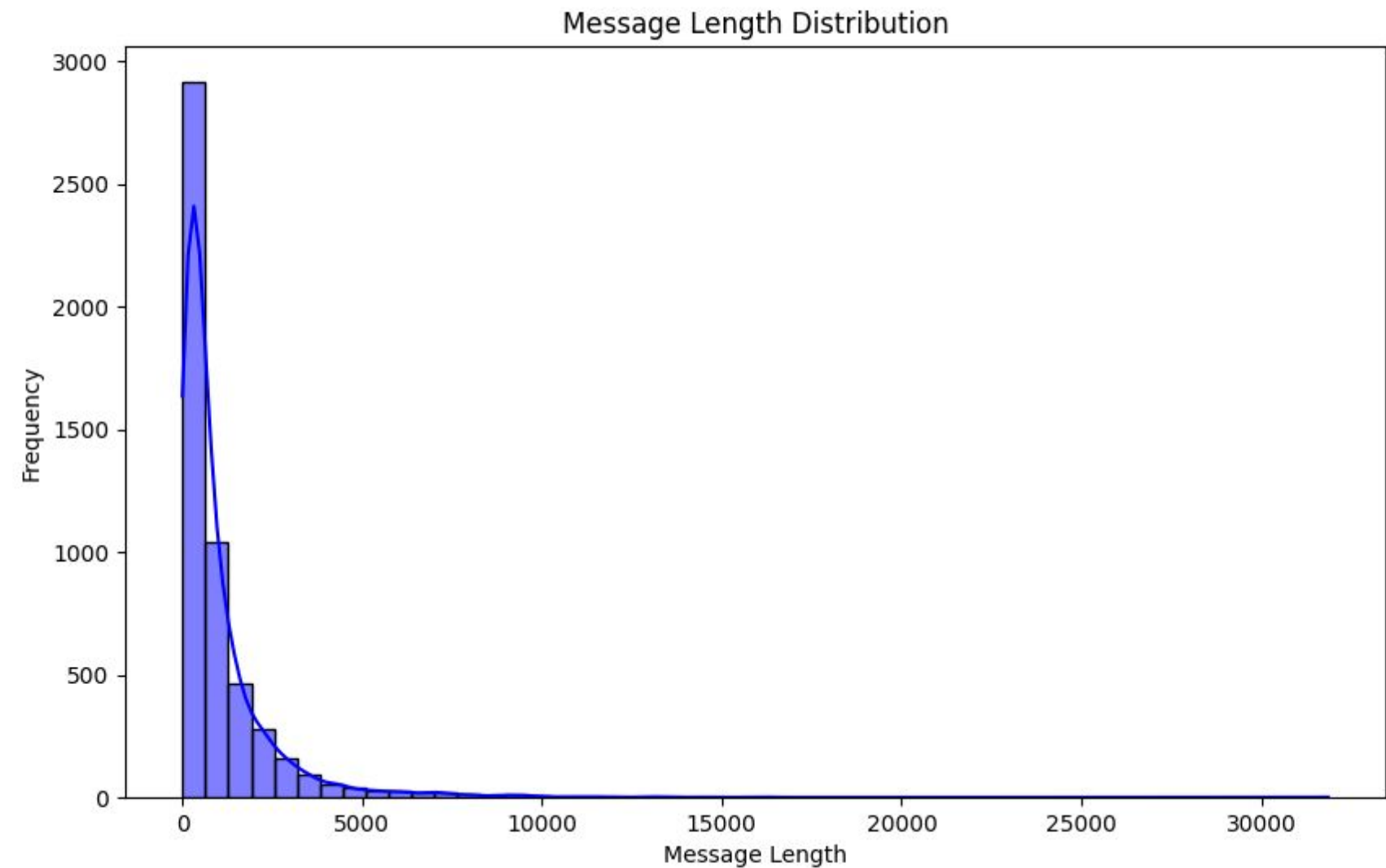
- Completed data extraction from Enron dataset
- Successfully cleaned and preprocessed enron1 data (there are 6)
 - Extracted messages from zip file and loaded ham/spam folders
 - Removed special characters, converted to lowercase
 - Eliminated stopwords using NLTK library
 - Created clean message column and saved to CSV
- Dataset is now properly formatted and labeled
- Performed data analysis:
 - Generated message length statistics and distributions
 - Analyzed class distribution between ham and spam
 - Created visualizations using matplotlib and seaborn
- Implemented feature engineering:
 - Added message length and word count features
 - Calculated punctuation counts and capital letter ratios
 - Identified spam keywords using TF-IDF vectorization
 - Created spam keyword frequency features

The distribution of emails in the dataset: approximately 3,500 Ham (legitimate) emails shown in blue versus 1,500 Spam emails shown in green.

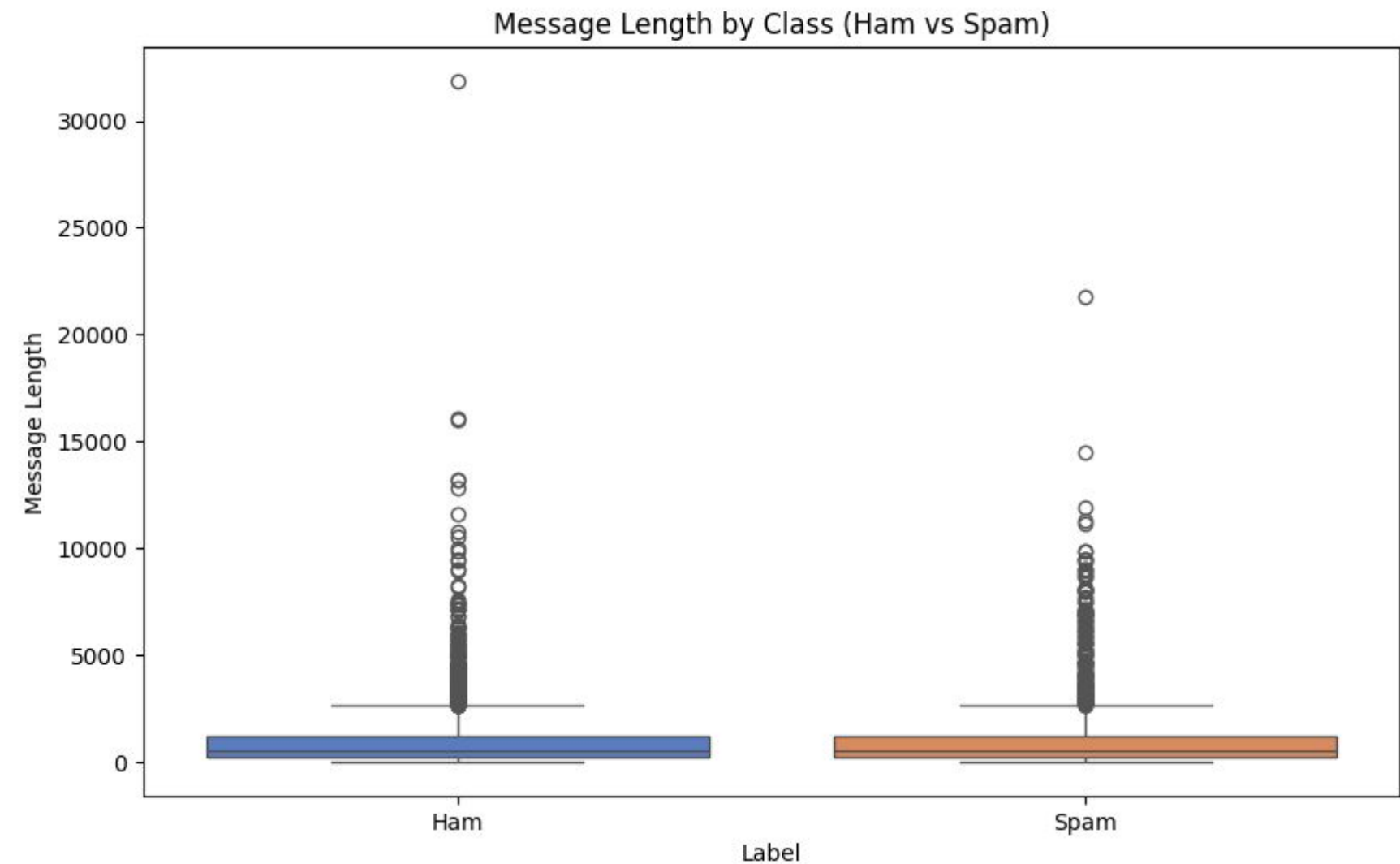


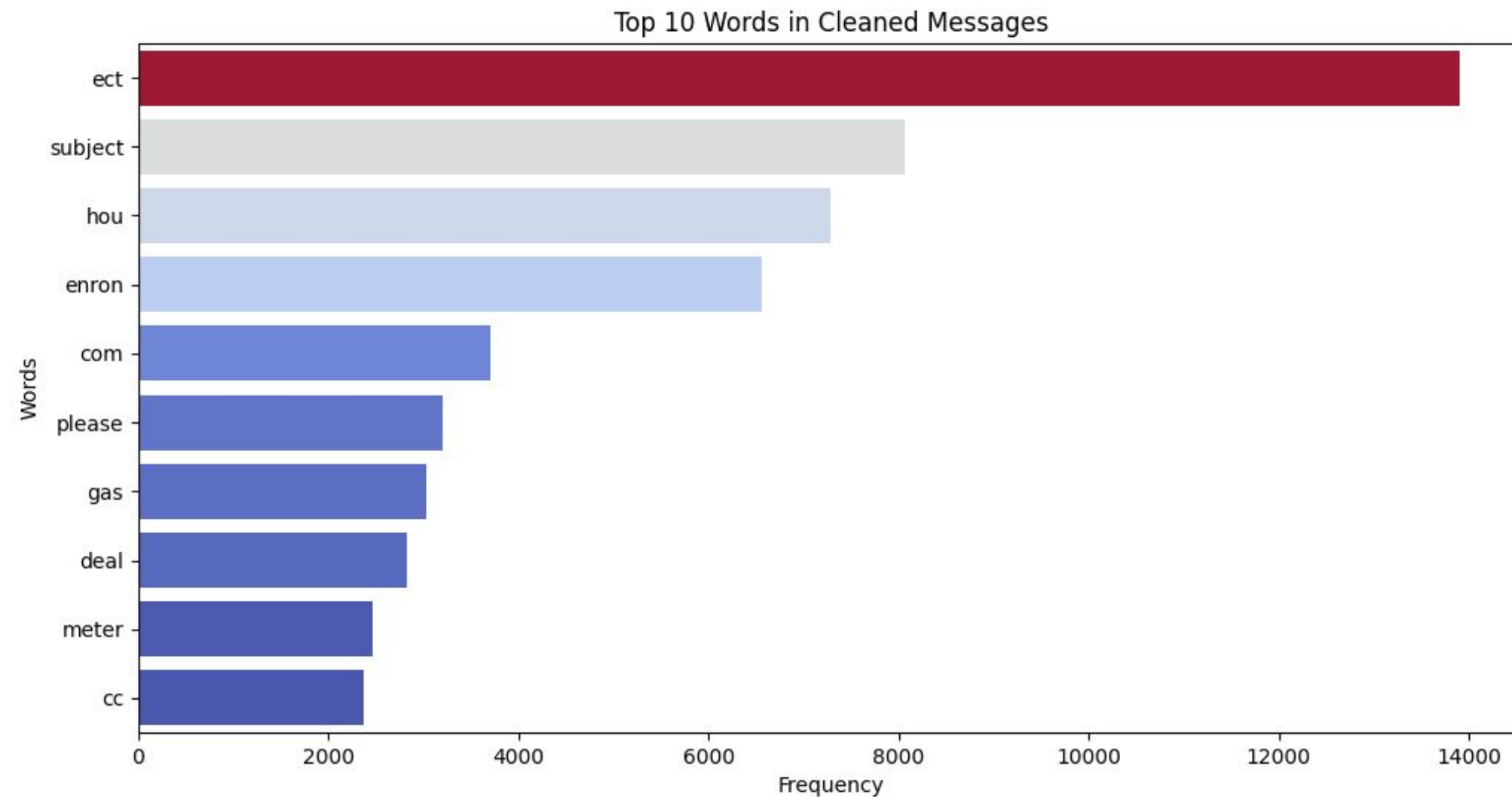
A histogram displaying the distribution of message lengths in the email dataset.

There's a peak frequency of around 2,800 characters at the shorter lengths, and the distribution gradually tails off towards longer message lengths up to 30,000 characters.



Boxplot visualization shows the distribution of message lengths for ham (legitimate) and spam emails. The plot reveals that spam messages tend to be shorter in length compared to ham messages.





The bar chart displays the top 10 most frequently occurring words in the cleaned email messages. The word "ect" appears most frequently with around 14,000 occurrences, followed by "subject" and "hou" with approximately 7,000-8,000 occurrences.

Engineered Features:

	Message_Length	Word_Count	Punctuation_Count	Capital_Letter_Ratio	Spam_Keyword_Count
0	2597	247	147	0.000385	3
1	549	30	38	0.001821	1
2	103	11	8	0.009709	1
3	530	44	81	0.001887	1
4	1391	139	136	0.000719	3

Engineered Features

- Message Length: Length of the message text.
- Word Count: Number of words in the cleaned message.
- Punctuation Count: Total punctuation marks in the original message.
- Capital Letter Ratio: Proportion of capital letters to total characters.
- Spam Keyword Count: Spam keywords identified and calculated their counts

Thank you!

Do you have any questions?